

METHODOLOGY FRAMEWORK

Adapting Layouts through Visual Analysis and Evolutionary Logic

Two Optimisation Goals

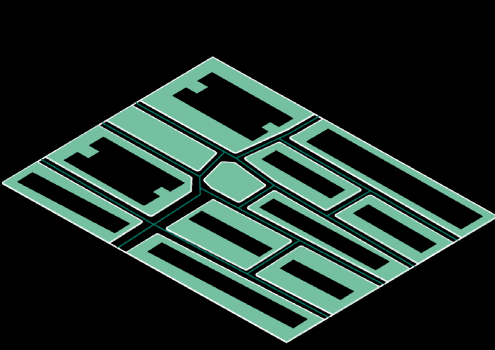
Green Space has been proved to be beneficial to human health especially to the elderly. Meanwhile, there is an increasing number of studies showing that the green visibility impacts the health of elderly greatly. Therefore, according to this, the project chose a residential area located in Guangdong Province, China as the research target.

This project is trying to promote data-driven and user-oriented urban design, particularly for aging societies, where green space configurations are no longer arbitrarily designed but intelligently optimized for well-being, inclusiveness, and long-term livability.

Stage 1: Site Analysis and Data Processing

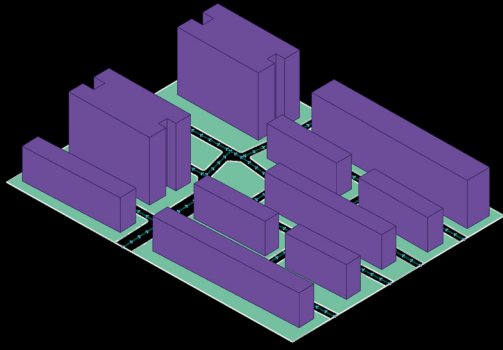
Green Boundaries Processing

Building Footprint Exclusion



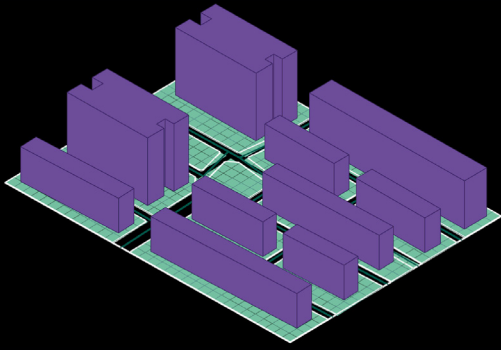
Valid Green Area

Paths' Sampling Points Generation (5m intervals)



Divide Paths' Distance

Valid Green Space Area Conversion to 5*5 Grid



Convert Surfaces into a 5*5 Grid

Stage 2: Grid Representation and Modelling

Greate Grid Units (gridRects)

Calcalater Centers Points (gridCenters)

Use Boolean Array to Represent Green Space Status (gridStates: True: Green Space False:Non-Green Space)

Distributed across All Zones Exclude the GreenAreas under Buildings

Generate Initial Population (Random Boolean Arrays)

Stage 3: Multi-Objective Optimization (NSGA-II)

Objective Function 1: Green Visibility Calculation

$$F_{GV} = 1/(n \cdot r) \sum_{i=1}^n \sum_{k=1}^r V(p_i, \theta_k)$$
$$V(p_i, \theta_k) = [G(p_i, \theta_k) \wedge \neg B(p_i, \theta_k)]$$

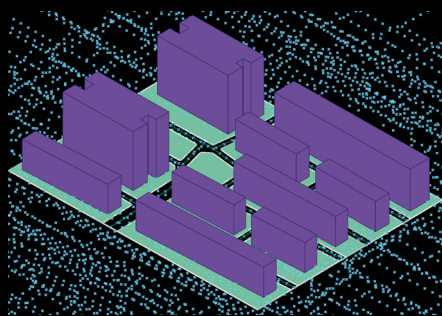
- F_{GV} = normalized green view index, ranging from [0, 1]
- n : number of sampling points
- r = number of rays emitted from each sampling point
- p_i : position of the i -th sampling point
- θ_k : angle of the k -th ray, $\theta_k = 2\pi k / r$
- $G(p_i, \theta_k)$ = whether the ray intersects with green space
- $B(p_i, \theta_k)$ = whether a building blocks the green space in front of the ray
- $[\cdot]$ = inverson bracket, which equals 1 if the condition inside is true, otherwise 0

Cast Rays from Sampling Points

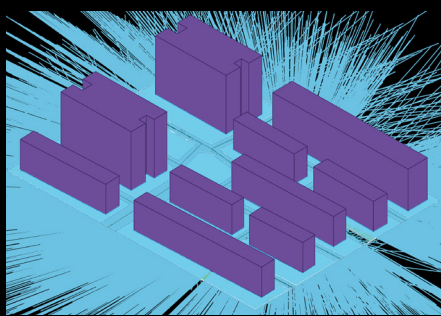
Ray Length 90-120 units

Detect Ray-Green Intersection

Building Obstruction Detection



Generate Isovist Points



Green Visibility Lines

Objective Function 2: Green Area Calculation

Actual Green Space Area Calculation

$$A_G = c^2 \cdot \sum_{i=1}^m g_i$$

Area Suitability Calculation

$$F_A = |A_G - A_T| + P_D$$

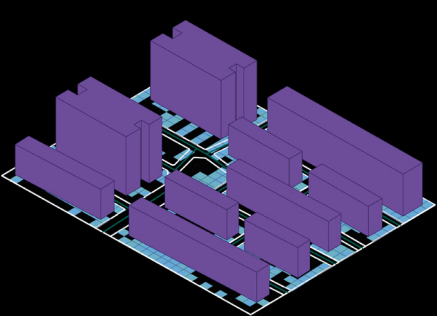
- A_G = total actual green space area
- c : size of each grid cells (cellSize)
- m = total number of grid cells
- g_i : status of the i -th grid cell (1 indicates green space, 0 indicates non-green space)
- F_A : area suitability value (the smaller, the better)
- A_T = target green space area (target area)
- P_D = penalty value for spatial distribution
- $| \cdot |$ = absolute value operator

Calculate Total Green Area

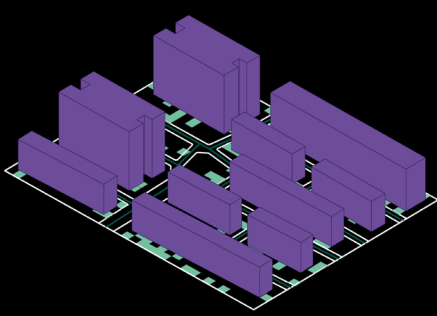
Compare with Target Area

Calculate Area Difference

Green Space Distribution Uniformity



Potential Green Area



Optimized Green Area

Pareto Optimal Solution Set: Maximize Green Visibility & Minimize Green Area Difference

ADAPTATION PROCESS

Algorithmic Thinking --- From Theory to Practice

Environmental Adaptation Mechanism

Spatial Recognition and Constraints

Building Adaptation: Automatically identify areas relative to building footprints.
Path Sampling: Optimize sampling point generation along walking paths.
Regional Adaptation: Automatically recognize and differentiate multiple zones.

Green Visibility Evaluation

Raycasting Simulation: Simulate human vision through ray emission to detect green visibility.
Occlusion Consideration: Account for building occlusions impacting green view

Multi-objective Optimization Process

NSGA-II Algorithm Implementation

Population Initialization: Generate initial green layouts randomly.
Multi-objective Fitness: Calculate green view gain and green area variance.
Regional Balancing: Use penalty factors to ensure green space distribution balance.
Non-dominated Sorting: Rank solutions based on Pareto optimality.

Trade-off Solution Selection

Calculate distance to ideal objectives and select a well-balanced solution.

Green Visibility Evaluation Mechanism

Green view evaluation is the core adaptation mechanism, ensuring optimization results match human perceptual experiences.
Each sampling point emits rays to assess visibility and building occlusion impacts, providing a perceptual-based green view ratio.

F_{GV} = Normalized green view index, ranging from [0, 1]

Green Space Layout Adaptation

Through iterative optimization, the green space layout gradually adapts to the spatial needs of the community environment.
From the initial random distribution to the final optimized state, the layout evolves into a more balanced pattern along pedestrian paths.
The system dynamically balances two competing objectives—green view improvement and area equality—meeting both perceptual experience needs and planning requirements.
Each region receives a reasonable allocation of green spaces, avoiding uneven distribution issues.

